

Apostol Chapter 1 - Exercise 6

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If $(a, b) = 1$ and if $d|(a + b)$, then $(a, d) = (b, d) = 1$.

Suppose that $(a, b) = 1$ and that $d|(a + b)$. Then, $a + b = dq$, for some integer q .

Obviously, 1 divides a, b and d .

Let e be a common divisor of a and d . That is, $a = eq_1$ and $d = eq_2$, for some integers q_1, q_2 . The follow holds,

$$\begin{aligned} dq = a + b &\Leftrightarrow eq_2q = eq_1 + b \\ &\Leftrightarrow b = e(q_2q - q_1), \end{aligned}$$

that is, $e|b$. Since $e|a$ and $e|b$, by definition, $e|(a, b)$, but $(a, b) = 1$. Thus, $e|1$. Therefore, $(a, d) = 1$.

A similar argument holds for b and d . Indeed, let f be a common divisor of b and d . Then, $b = fq_3$ and $d = fq_4$, for some integers q_3 and q_4 . We have the following,

$$\begin{aligned} dq = a + b &\Leftrightarrow fq_4q = a + fq_3 \\ &\Leftrightarrow a = f(q_4q - q_3), \end{aligned}$$

that is, $f|a$. Since $f|a$ and $f|b$, by definition, $f|(a, b)$, but $(a, b) = 1$. Thus, $f|1$. Therefore, $(b, d) = 1$.